POWER BI · DATA ANALYTICS PROJECT

HR Analytics –

Employee Performance & Payroll Dashboard

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Attrition · Payroll · Performance · Engagement

Industry Selection

Human Resources (HR) Analytics

Every company runs on people. HR analytics turns employee data into clear answers.

People are the biggest asset

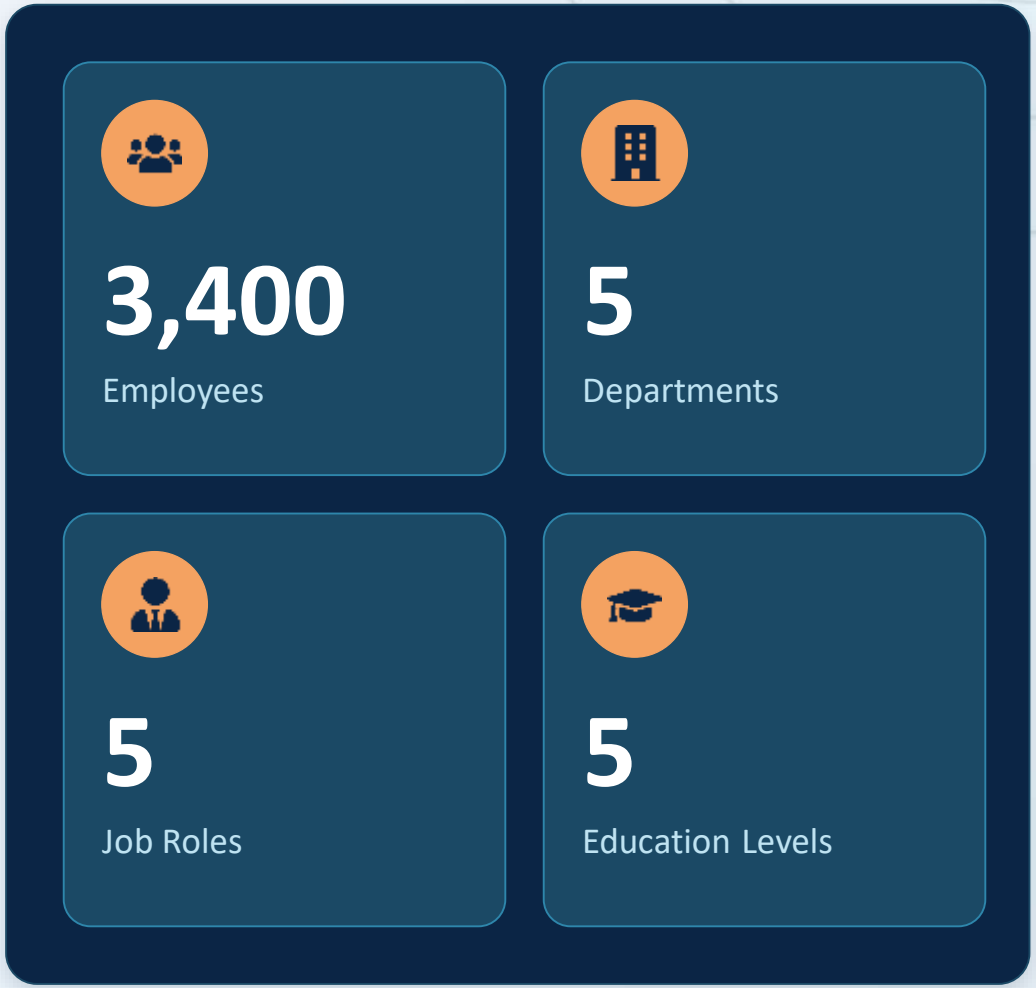
Hiring, paying and keeping staff costs a lot.

Attrition is expensive

When people leave, skills and money leave too.

Data guides decisions

Dashboards show HR where to act.



Problem Statement

The company has HR data in a flat CSV file but no clear picture. We need to find out who is leaving, where the payroll goes, and how employees perform.

Who is leaving?

Attrition by role, age, tenure and overtime

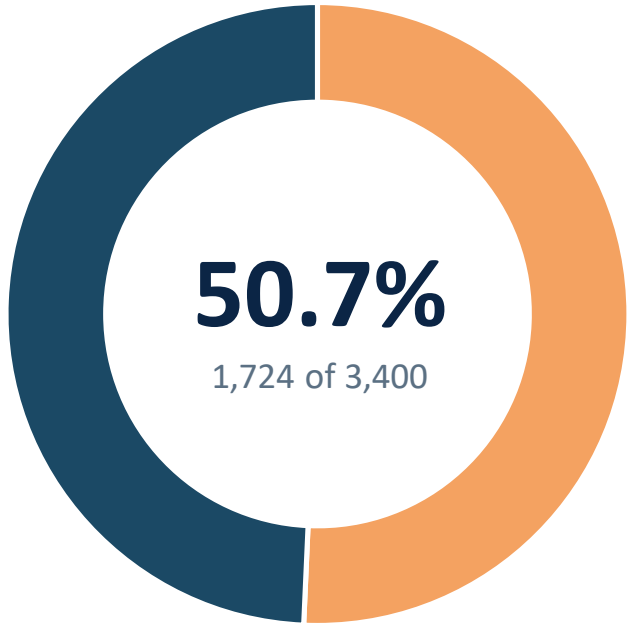
Where does the money go?

Payroll by department, role and education

How are employees doing?

Performance, satisfaction and work-life balance

Employees who left



Dataset Collection (CSV)

kaggle

HR Employee Dataset

Source: Kaggle · Format: CSV

3,400	20	1
Rows	Columns	CSV file

Open dataset on Kaggle

kaggle.com/datasets/justinlimyin/hr-employee-dataset

Employee

Age, Gender
Education level
Marital status
Companies worked

Job

Department, Job role
Monthly income
Years at company
Years in role

Performance

Performance rating
Job satisfaction
Work-life balance
Training hours

Exit & Work

Attrition (Yes/No)
Overtime
Distance from home
Stock option level

Raw Dataset Backup

Keep the original safe

The raw file is never edited.

Compare before and after

Raw has 20 columns, clean has 28.

Repeat the work anytime

Cleaning can always start again.



Dataset / folder

HR_Dataset_raw.csv	304 KB	**Original**
HR_Employee_Clean.csv	477 KB	**+8 columns**
Dim_Department.csv	1 KB	**Lookup**
Dim_JobRole.csv	1 KB	**Lookup**
Dim_Education.csv	1 KB	**Lookup**

Power BI ETL – Power Query

1



Get Data

Home → Get data → Text/CSV. Pick HR_Employee_Clean.csv.

2



Transform

Open Power Query. Rename to Fact_Employee. Check data types.

3



Add lookups

Load Dim_Department, Dim_JobRole and Dim_Education.

4



Close & Apply

Load all tables into the Power BI model.



Result: 4 clean tables ready in Power BI: Fact_Employee, Dim_Department, Dim_JobRole and Dim_Education.

Data Cleaning & Transformation

Cleaned

- Data types set
- Text trimmed
- Duplicates: 0 found
- Blank values: 0 found

New columns added

- Age_Group
- Income_Band
- Tenure_Band
- Annual_Income
- Data_Quality_Flag
- + Sort columns

Data quality flag

Odd rows were kept and flagged, not deleted.

957	608	32
Tenure exceeds working age	Role tenure exceeds company tenure	Both problems

DAX & KPI Creation

~35

DAX measures

Check: Total Employees = 3,400

3,400
Total Employees

50.7%
Attrition Rate

1,74,26,235
Monthly Payroll

5,125
Avg Monthly Income

41.0%
High Performer %

48.2%
Overtime %

Attrition Rate

Employees who left ÷ Total employees

Monthly Payroll

Sum of every monthly income

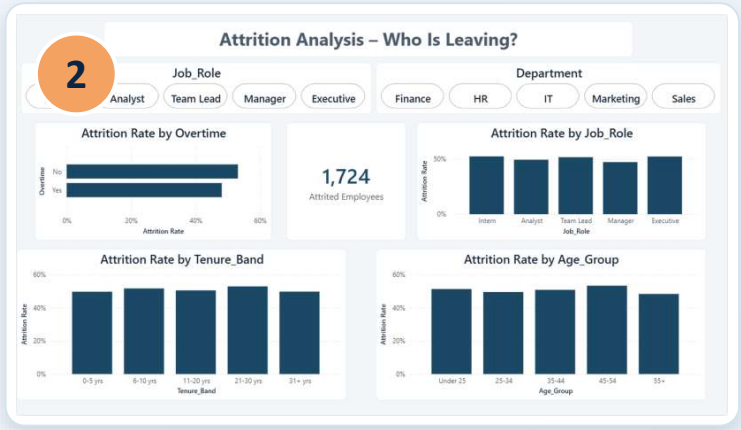
High Performer %

Rating 4 or 5 ÷ Total employees

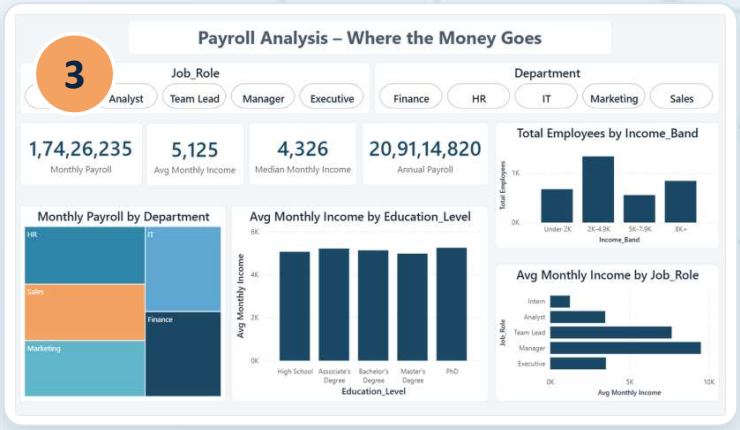
Dashboard Development – 5 Pages



Executive Overview



Attrition Analysis



Payroll Analysis



Performance & Engagement



Employee Details

In every page

- Job Role slicer
- Department slicer
- Cross-filtering visuals
- Custom colour theme

Dashboard 1: Executive Overview



Six KPI cards

3,400 staff · 50.7% attrition

Department mix

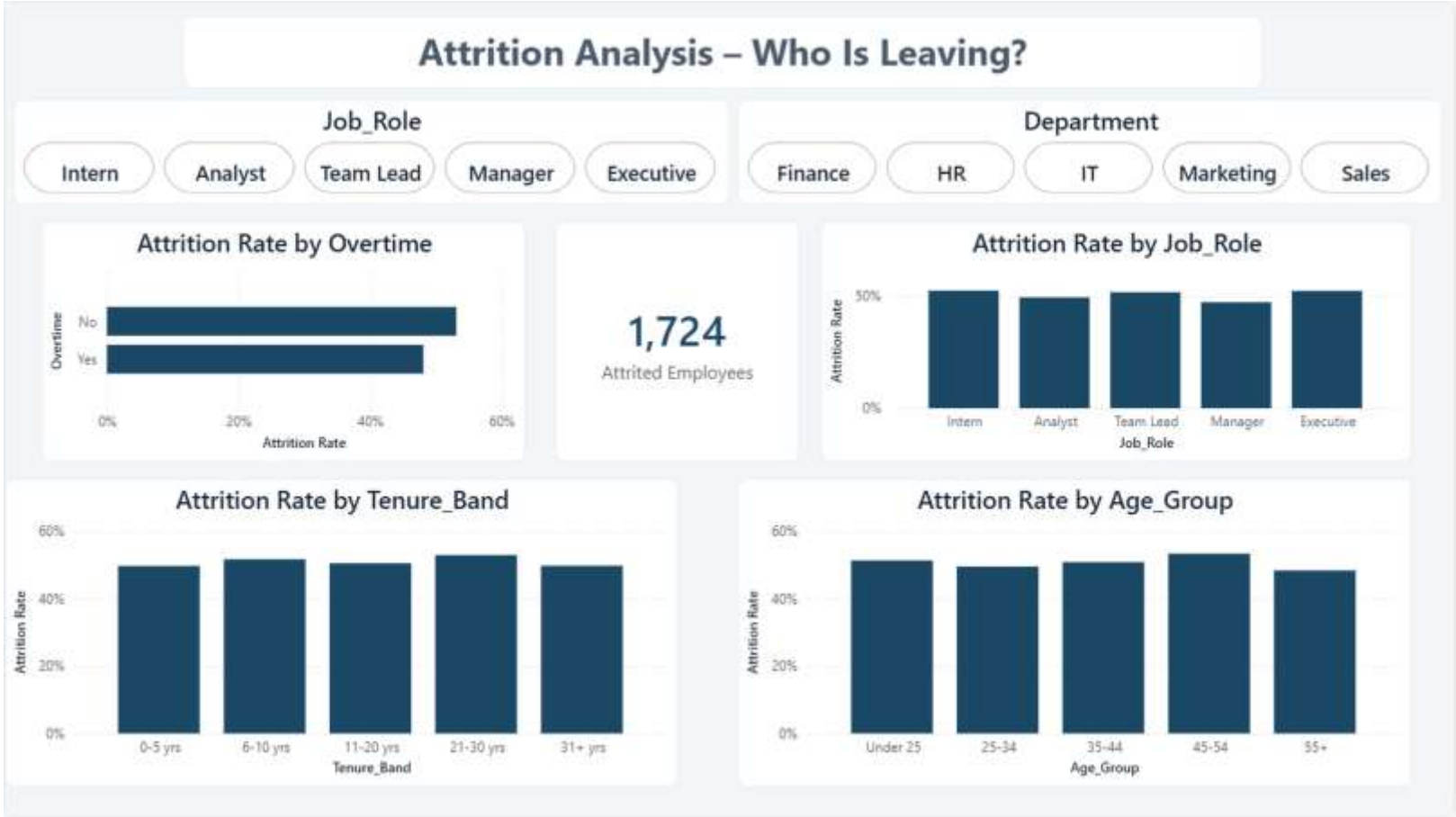
About 650–700 per department

Slicers

Job Role and Department filter all

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Dashboard 2: Attrition Analysis



1,724 employees left

Half of all 3,400 staff

Overtime

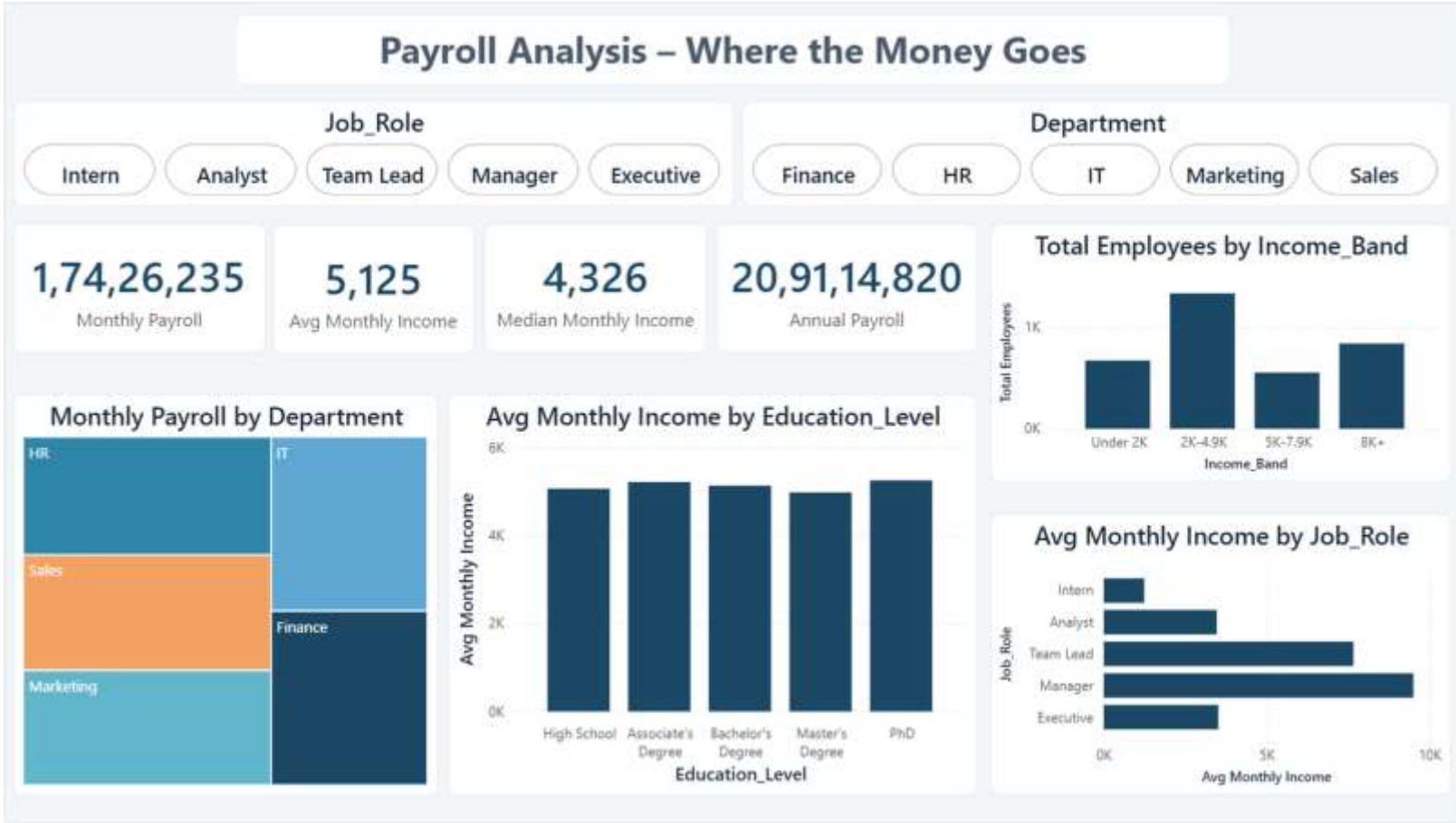
No overtime 53.1% vs 48.1%

Role, tenure, age

All near 50%, no clear driver

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Dashboard 3: Payroll Analysis



Payroll totals

Monthly 1,74,26,235 · Annual 20,91,14,820

Average vs median

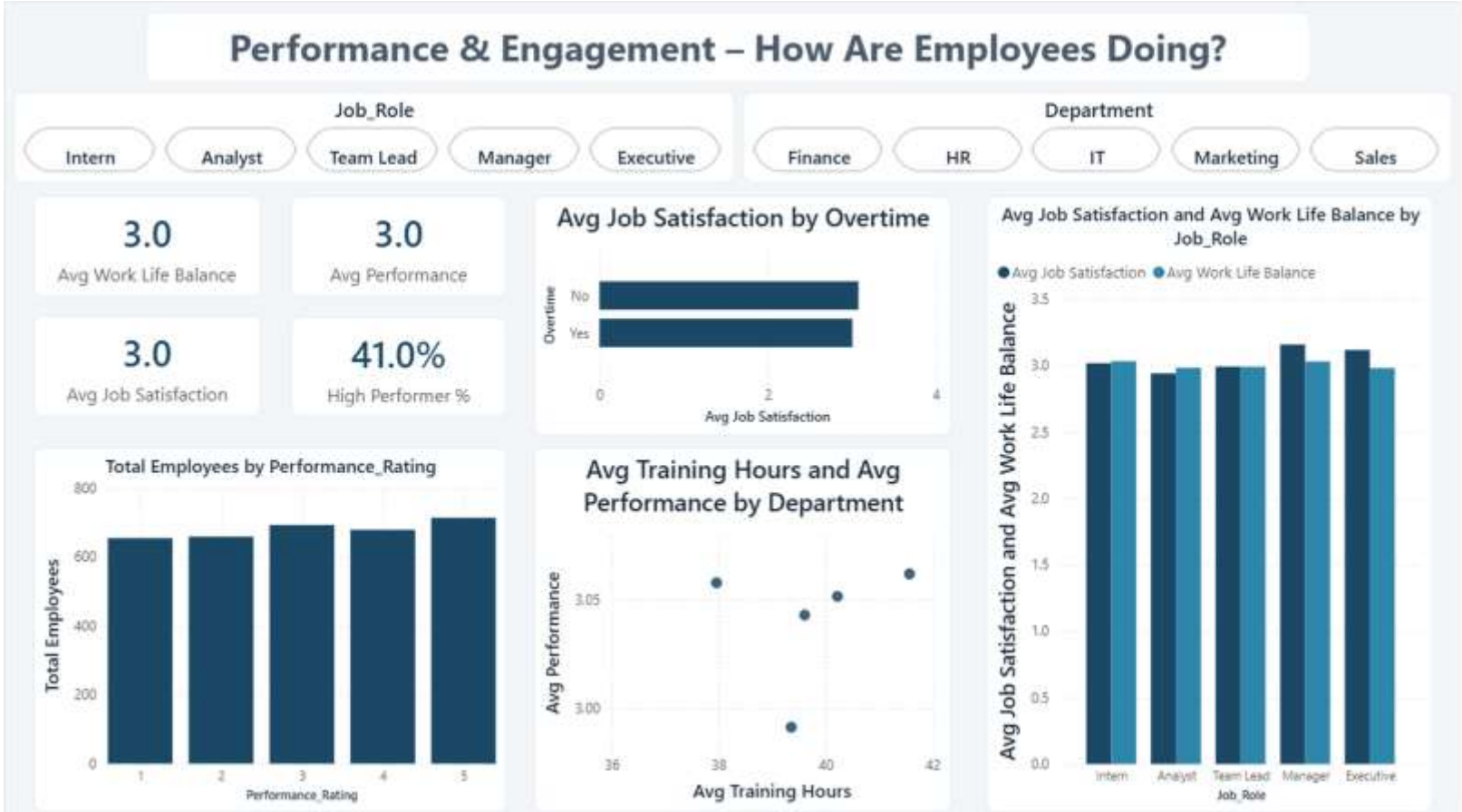
5,125 average · 4,326 median

By job role

Manager 9,505 · Intern 1,243

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Dashboard 4: Performance & Engagement



Average scores

Satisfaction, balance, performance ≈ 3.0

High performers

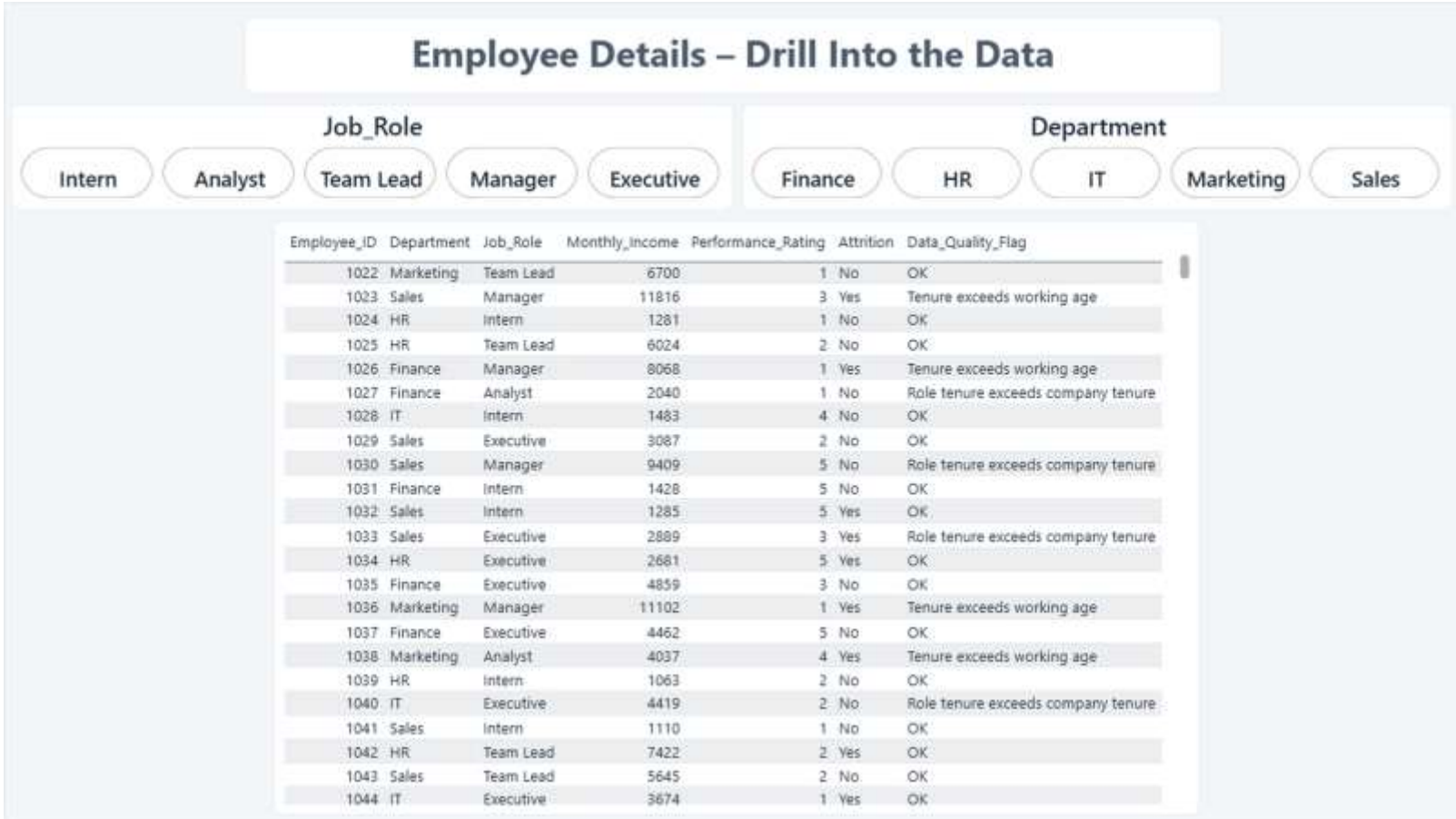
41.0% rated 4 or 5

Training vs performance

No strong link by department

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Dashboard 5: Employee Details



Employee_ID	Department	Job_Role	Monthly_Income	Performance_Rating	Attrition	Data_Quality_Flag
1022	Marketing	Team Lead	6700	1	No	OK
1023	Sales	Manager	11816	3	Yes	Tenure exceeds working age
1024	HR	Intern	1281	1	No	OK
1025	HR	Team Lead	6024	2	No	OK
1026	Finance	Manager	8068	1	Yes	Tenure exceeds working age
1027	Finance	Analyst	2040	1	No	Role tenure exceeds company tenure
1028	IT	Intern	1483	4	No	OK
1029	Sales	Executive	3087	2	No	OK
1030	Sales	Manager	9409	5	No	Role tenure exceeds company tenure
1031	Finance	Intern	1428	5	No	OK
1032	Sales	Intern	1285	5	Yes	OK
1033	Sales	Executive	2889	3	Yes	Role tenure exceeds company tenure
1034	HR	Executive	2681	5	Yes	OK
1035	Finance	Executive	4859	3	No	OK
1036	Marketing	Manager	11102	1	Yes	Tenure exceeds working age
1037	Finance	Executive	4462	5	No	OK
1038	Marketing	Analyst	4037	4	Yes	Tenure exceeds working age
1039	HR	Intern	1063	2	No	OK
1040	IT	Executive	4419	2	No	Role tenure exceeds company tenure
1041	Sales	Intern	1110	1	No	OK
1042	HR	Team Lead	7422	2	Yes	OK
1043	Sales	Team Lead	5645	2	No	OK
1044	IT	Executive	3674	1	Yes	OK

Employee table

One row per employee

Data quality flag

Shows rows that need care

Drill down

Filter by role or department

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Business Insights

Attrition is about 50% everywhere

Same in every department, role, age and tenure group.

No overtime left slightly more

53.1% without overtime vs 48.1% with overtime.

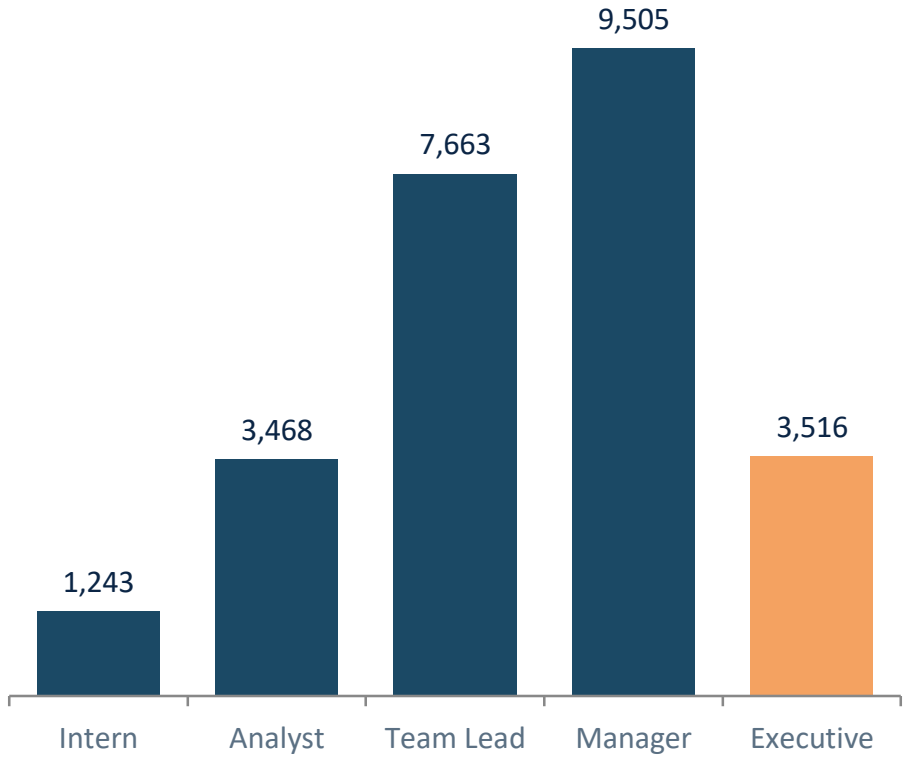
Executives earn less than Managers

3,516 vs 9,505 per month. This is unusual.

Almost no link between scores

Performance, income and satisfaction barely relate.

Average monthly income by role



Note: the data looks synthetic, so this is a portfolio project, not a real HR study.

Publish to Power BI

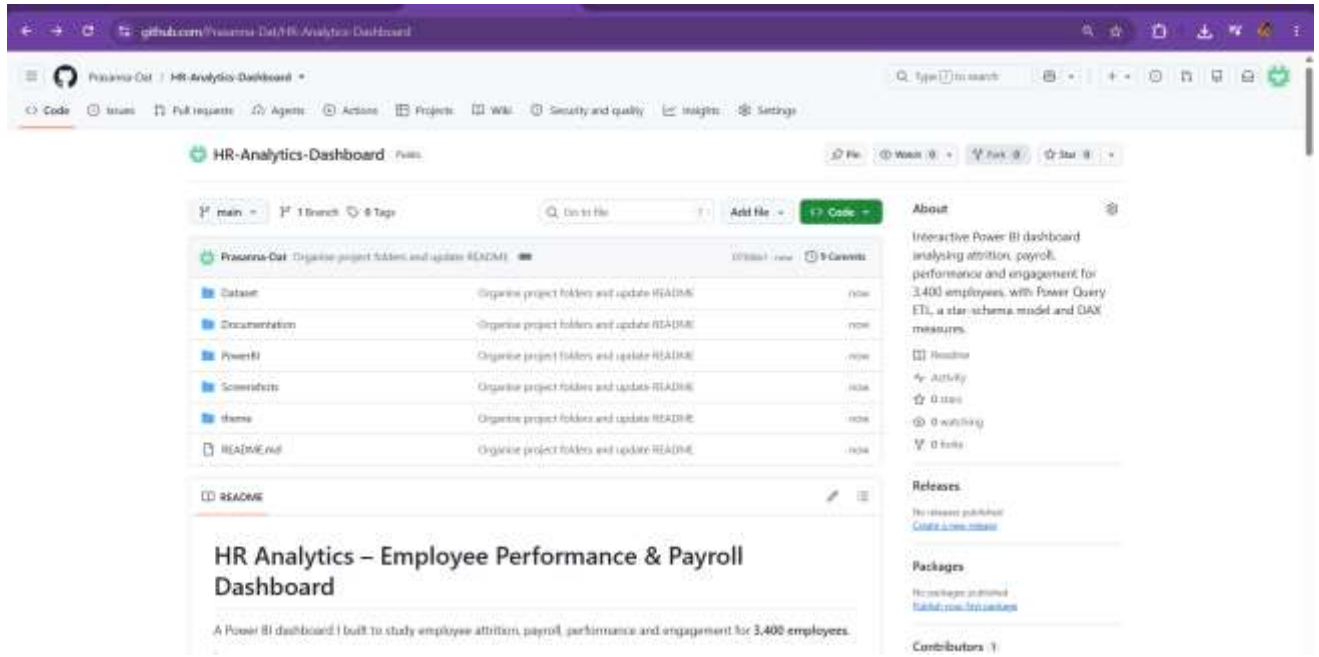
1. **Save the report**
   HR_Analytics_Dashboard.pbix
2. **Publish**
   Home → Publish, then sign in
3. **Choose a workspace**
   Pick My workspace or a team one
4. **Open and share**
   Open the report online, share the link



Power BI Service

- Interactive slicers
- Cross-filtering visuals
- Shareable link

GitHub Documentation



github.com/Prasanna-Dat/HR-Analytics-Dashboard

Public repository · Click to open

Repository structure

Dataset/
Raw, clean and lookup CSVs

Documentation/
Data dictionary, project report

PowerBI/
HR_Analytics_Dashboard.pbix

Screenshots/
5 dashboard page images

theme/
HR_Theme.json colours

README.md
Findings, how to open, limits



Thank You

Questions and feedback are welcome

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Center ID: TNJPM

GitHub Repo

kaggle Kaggle Dataset

View Dashboards

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